its effort into fighting, even less to gain by putting its effort into caring. Exactly the opposite will be true of the *B*s as the generations go by. The key idea here is that a small initial difference between the sexes can be self-enhancing: selection can start with an initial, slight difference and make it grow larger and larger, until the *A*s become what we now call males, the *B*s what we now call females. The initial difference can be small enough to arise at random. After all, the starting conditions of the two sexes are unlikely to be exactly identical.

As you will notice, this is rather like the theory, originating with Parker, Baker, and Smith and discussed on page 185, of the early separation of primitive gametes into sperms and eggs. The argument just given is more general. The separation into sperms and eggs is only one aspect of a more basic separation of sexual roles. Instead of treating the sperm–egg separation as primary, and tracing all the characteristic attributes of males and females back to it, we now have an argument that explains the sperm–egg separation and other aspects all in the same way. We have to assume only that there are two sexes who have to mate with each other; we need know nothing more about those sexes. Starting from this minimal assumption, we positively expect that, however equal the two sexes may be at the start, they will diverge into two sexes specializing in opposite and complementary reproductive techniques. The separation between sperms and eggs is a symptom of this more general separation, not the cause of it.

***p. 196 Let us take Maynard Smith's method of analysing aggressive contests, and apply it to sex.***

This idea of trying to find an evolutionarily stable mix of strategies within one sex, balanced by an evolutionarily stable mix of strategies in the other sex, has now been taken further by Maynard Smith himself and, independently but in a similar direction, by Alan Grafen and Richard Sibly. Grafen and Sibly's paper is the more technically advanced, Maynard Smith's the easier to explain in words. Briefly, he begins by considering two strategies, guard and desert, which can be adopted by either sex. As in my 'coy/fast and faithful/philanderer' model, the interesting question is: what combinations of strategies among males are stable against what combinations of strategies among females? The answer depends upon our assumption about the particular economic circumstances of the species. Interestingly, though, however much we vary the economic assumptions, we don't have a whole continuum of quantitatively varying stable outcomes. The model tends to home in one of only four stable outcomes. The four outcomes are named after animal species that exemplify them. There is the Duck (male deserts, female guards), the Stickleback (female deserts, male guards), the Fruit-fly (both desert), and the Gibbon (both guard).

And here is something even more interesting. Remember from Chapter 5 that ESS models can settle at either of two outcomes, both equally stable? Well, that is true of this Maynard Smith model, too. What is especially interesting is that particular pairs, as opposed to other pairs, of these outcomes are jointly stable under the same economic circumstances. For instance, under one range of circumstances, both Duck and Stickleback are stable. Which of the two actually arises depends upon luck or, more precisely, upon accidents of evolutionary history—initial conditions. Under another range of circumstances, both Gibbon and Fruit-fly are stable. Again, it is historical accident that determines which of the two occurs in any given species. But there are no circumstances in which Gibbon and Duck are jointly stable, no circumstances in which Duck and Fruit-fly are jointly stable. This 'stablemate' (to coin a double pun) analysis of congenial and uncongenial combinations of ESSs has interesting consequences for our reconstructions of evolutionary history. For instance, it leads us to expect that certain kinds of transitions between mating systems in evolutionary history will be probable, others improbable. Maynard Smith explores these historical networks in a brief survey of mating patterns throughout the animal kingdom, ending with the memorable rhetorical question: 'Why don't male mammals lactate?'